PAPER_751: THz Q-Scope Earth Core Resonance Signals — Channels 1–50

Author: Daniel T. Murphy
Framework: Universal Quantum Field Superconductive Framework (UQFF)
Session: 181 | v5.39
Date: 2026
CP4 Class: #335 — THzQScopeEarthCoreSig1to50Calculator

Abstract

A THz Q-Scope instrument monitoring Earth's core resonance detects 50 discrete signal channels between 1.2–1.3 THz. The UQFF framework interprets these signals through the [{U_m:SM_m}/[Ug1=UQFF_g+SM_g]^(SCm)] resonance operator, wherein a "THz hole" forms between two pseudo-monopole poles. This paper derives the peak power, effective voltage amplitude, and angular resonance frequency for the instrument's 1.246 Hz sampling window, and maps the 50-channel ensemble to the UQFF vacuum coupling hierarchy.

---

1. Introduction

Seismic and electromagnetic probes of Earth's deep interior reveal coherent oscillation modes at terahertz frequencies unexplained by standard core models. The THz Q-Scope instrument records 50 channels (dA = 6.205 A, ±3.102 A dynamic range) at a display sampling frequency of 1.246 Hz. The actual resonance lies at 1.2–1.3 THz — 12 orders of magnitude above the displayed sampling clock.

Within UQFF, the Earth's conducting outer core acts as a pair of pseudo-monopole boundaries separated by a "THz hole" in the Ug1 magnetic-dipole vacuum field. The observable [{U_m:SM_m}/[Ug1]^(SCm)] ratio governs which channels couple to the surface and which are attenuated below the superconductive threshold H_SCm ≈ 0.99.

---

2. Master Resonance Equation

```
P_THz(f) = [{U_m : SM_m} / [Ug1(r,t) + SM_g]^(SCm)]
           × V_eff² / Z_imp
           × Σ_{k=1}^{50} δ(f − f_k)
```

Where: - U_m = UQFF magnetic vacuum energy density - SM_m = Standard Model magnetic field contribution - Ug1 = UQFF_g + SM_g (total gravity+EM coupling at core radius) - SCm = superconductive modifier ≈ H_SCm = 0.99 - V_eff = RMS voltage amplitude = 0.35 V - Z_imp = instrument impedance = 50 Ω - f_k = individual channel centre frequency (k = 1…50) - δ(·) = Dirac delta selector

Angular Frequency

```
ω_THz = 2π × f_THz = 2π × 1.25×10¹² ≈ 7.854×10¹² rad/s
```

Peak Power

```
P_peak = V_eff² / Z_imp = (0.35)² / 50 = 0.00245 W = 2.45 mW
```

Effective Current

```
I_eff = V_eff / Z_imp = 0.35 / 50 = 7.0×10⁻³ A
dA (full-scale) = 6.205 A  (±3.102 A symmetric swing)
```

---

3. UQFF Vacuum Coupling Parameters

Parameter	Symbol	Value	Unit
Sampling frequency	f_samp	1.246	Hz
THz resonance	f_THz	1.25×10^{12}	Hz
Angular frequency	ω_THz	7.854×10^{12}	rad/s
Peak-to-peak voltage max	V_pp_max	1.00	V
Effective voltage	V_eff	0.35	V
Instrument impedance	Z_imp	50	Ω
Peak power	P_peak	2.45×10^{-3}	W
Full-scale current	dA	6.205	A
Channels	N_ch	50	—
UQFF vacuum density (UA)	ρ_UA	7.09×10^{-36}	kg/m^3
UQFF vacuum density (SCm)	ρ_SCm	7.09×10^{-37}	kg/m^3
Superconductive modifier	SCm	0.99	—

---

4. THz Hole Formation

The resonance arises from frequency cloistering between two pseudo-monopole poles at the inner core boundary ($r \approx 1.22\times10^6$ m) and the outer core–mantle boundary ($r \approx 3.48\times10^6$ m). In UQFF:

```
Ug1_core(r) = G·M_core/(r²) × (1 + μ_J × B_core²/(ρ_core × c²))
```

The superconductive term (1 − B/B_crit)^SCm suppresses non-resonant modes, leaving 50 coherent channels visible to surface instrumentation.

---

5. Conclusions

The THz Q-Scope framework identifies 50 Earth-core resonance channels at $\omega \approx 7.85\times10^{12}$ rad/s with 2.45 mW peak power per channel at 50 Ω impedance. The UQFF [{U_m:SM_m}/Ug1^SCm] coupling ratio explains both the channel selection and the large amplitude ratio (dA = 6.205 A) relative to the milliwatt power scale. PAPER_751, CP4 class #335. v5.39.